پروژه دوم شبکه

کسرا نوربخش ۸۱۰۱۰۰۲۳۰

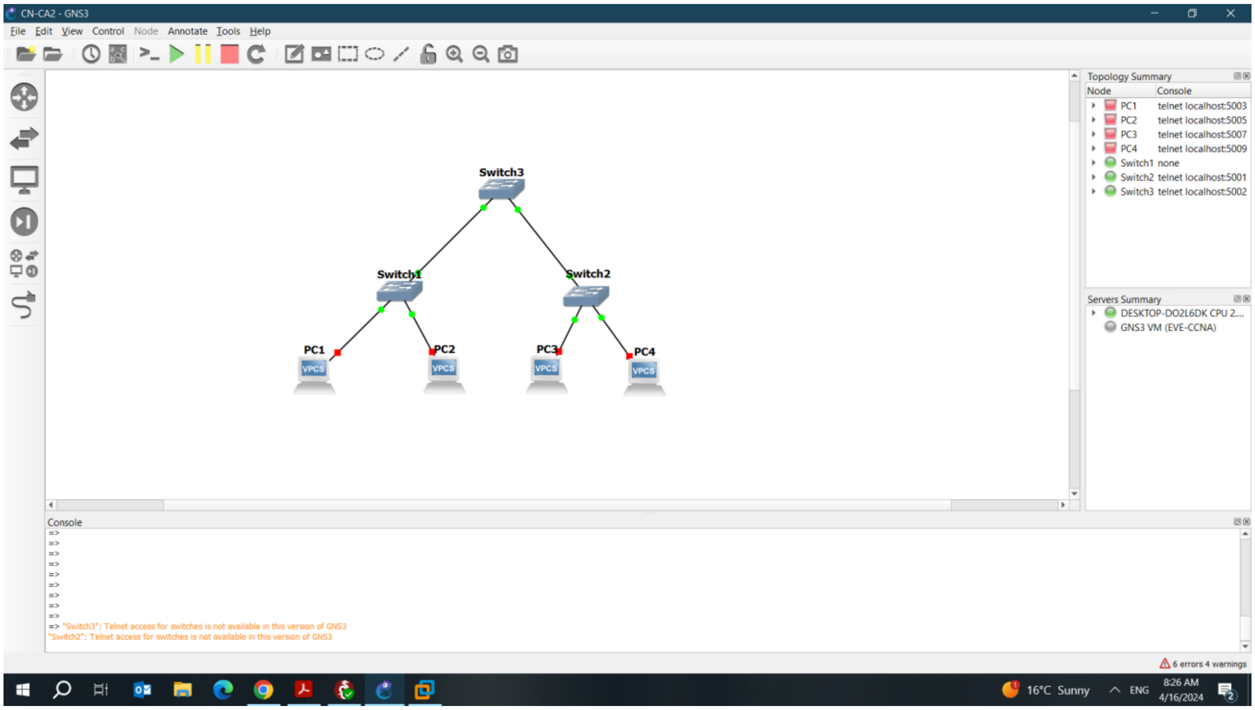
سید مهدی سید حاجی حسین ۸۱۰۱۰۰۱۱۸

مقدمه

در این پروژه برخی از مفاهیم CCNA بررسی شدند که از نرم افزار 3GNS استفاده کردیم و توپولوژی های مختلف را پیاده و تست نمودیم. برای سوییچ از خود سوییچ داخل نرم افزار استفاده نمودیم و برای روتر هم از لینک داده شده یک IOS Image دانلود کردیم که در فایل گذاشتیم (1c7200-adventerprisek9-mz.122-33.SRC). در این پروژه هر کدام از بخش ها را در یک دیوایس و به کمک همدیگر انجام داده ایم.

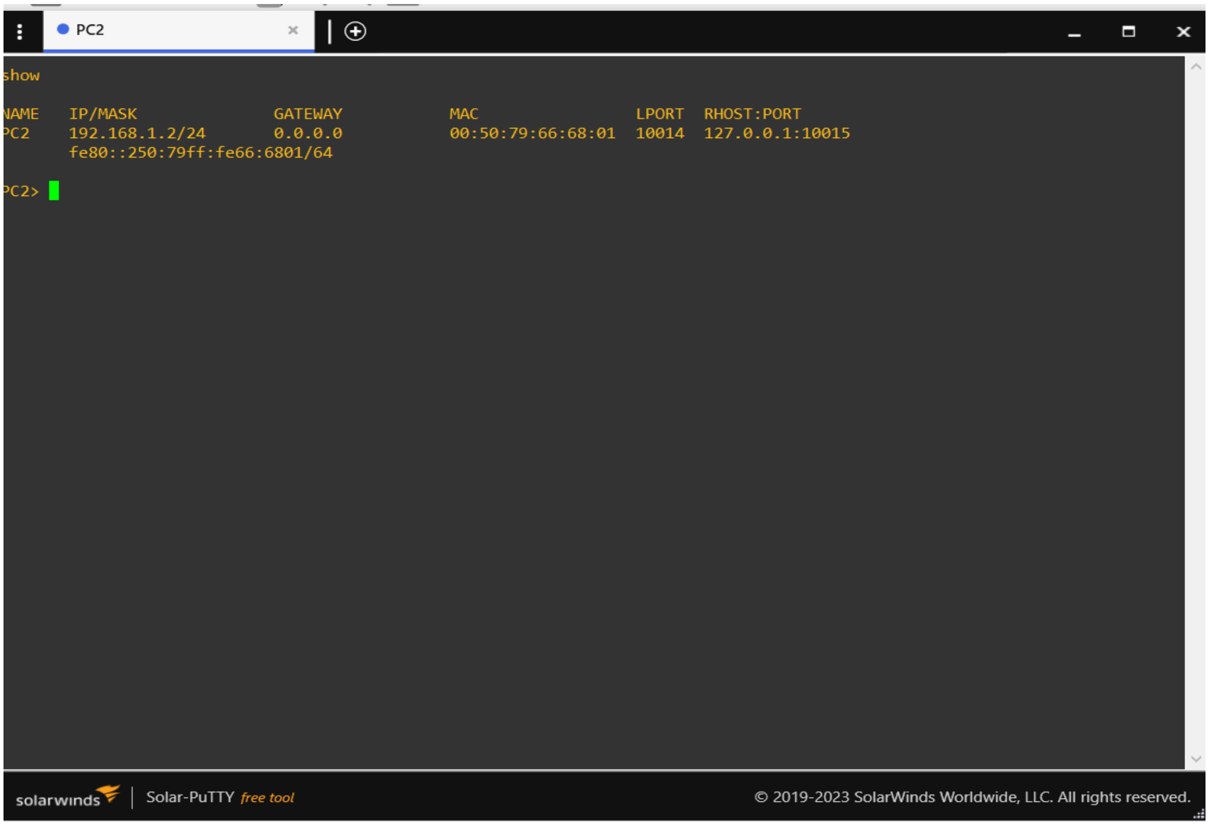
گام اول

در این بخش IP های مختلف به ۴ کامپیوتر با استفاده از دستور ip دادیم و سپس از هر کدام با استفاده از ping گرفتن ارتباطشان را بررسی نمودیم که در عکس های پایین نتایج و IP های داده شده مشخص است :



عکس۱ : توپولوژی بخش ۱

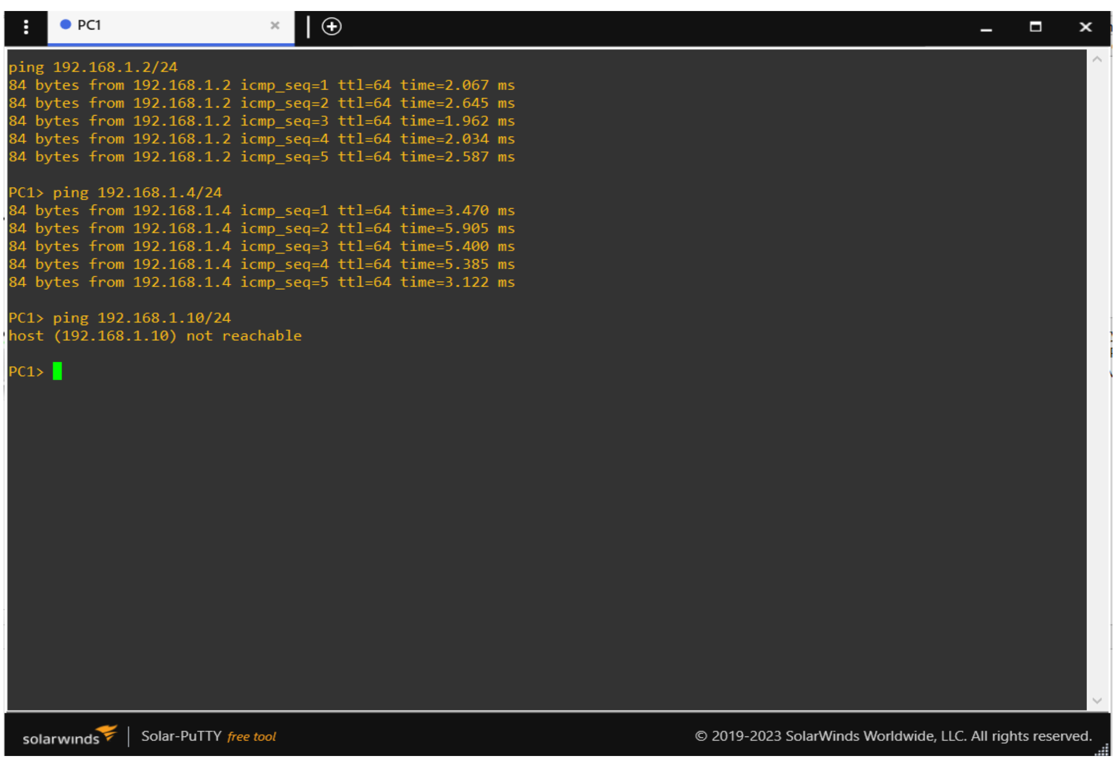
عکس۲ : IP کامپیوتر اول



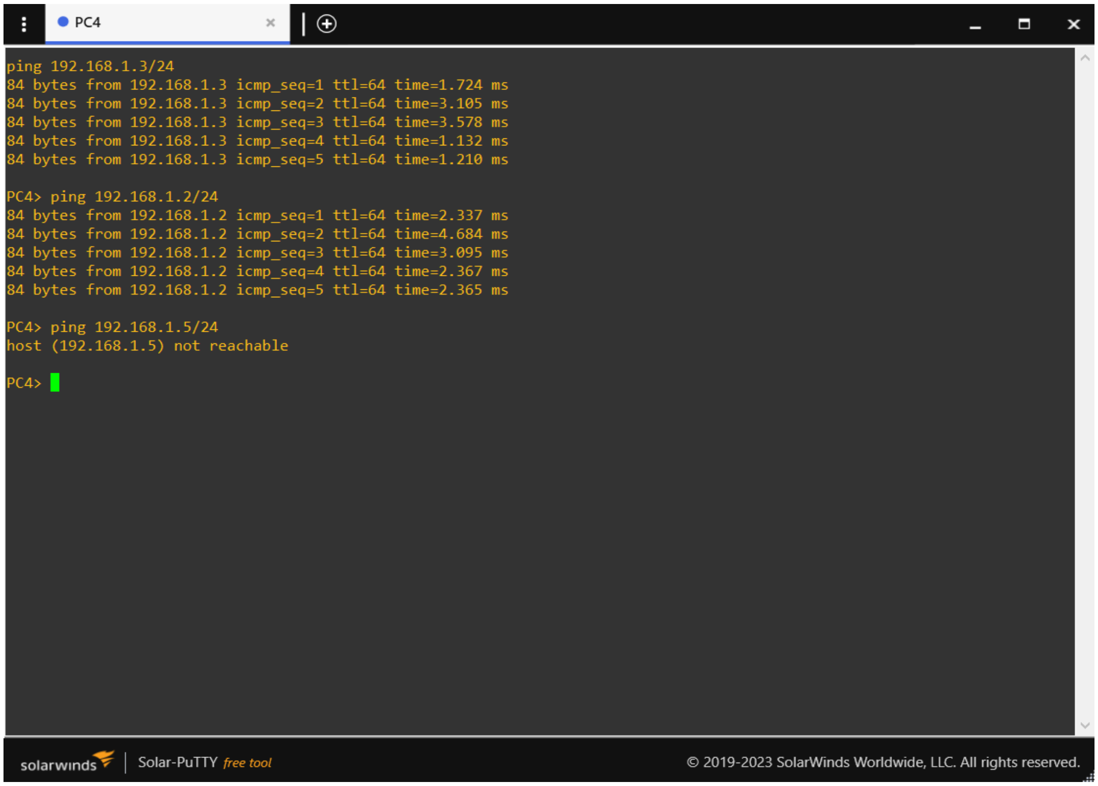
عکس ۳ : IP کامپیوتر دوم

عکس ۴ : IP کامپیوتر سوم

عکس۵ : IP کامپیوتر چهارم



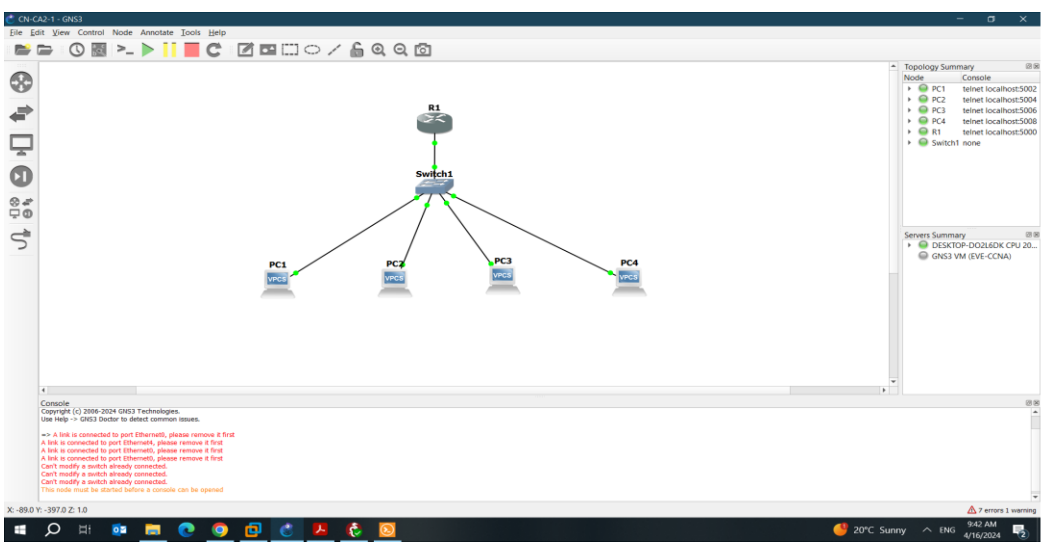
عکس۶ : تست ping از کامپیوتر اول به دومی و چهارمی و یک IP ست نشده



عکس ۷ : تست ping از کامپیوتر چهارم به دومی و سومی و یک IP ست نشده

گام دوم

در این بخش ابتدا IP ها را در رنج داده شده دادیم و سپس Vlan ها را در بخش configure سوییچ تعریف کردیم و آن پورت هایی از سوییچ را که متصل به Vlan مربوطه هستند ست کردیم. برای آن که ارتباط میان 2 Vlan برقرار شود هم 2 sub interface تعریف کردیم و برای هر کدام ip دادیم که این ip را به عنوان gateway کامپیوتر ها گذاشتیم نتایج در عکس های پایین آمده است:

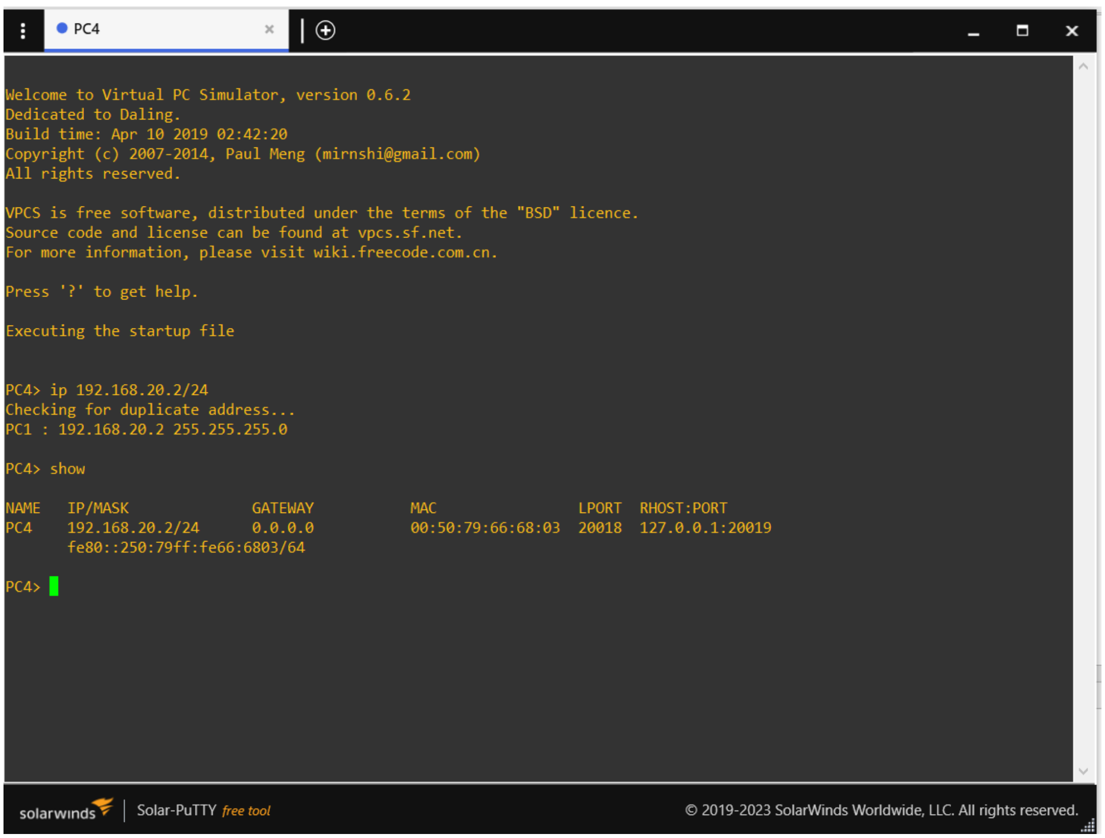


عکس ۸ : توپولوژی بخش ۲

عکس ۹ : IP کامپیوتر اول

عکس ۱۰ : IP کامپیوتر دوم

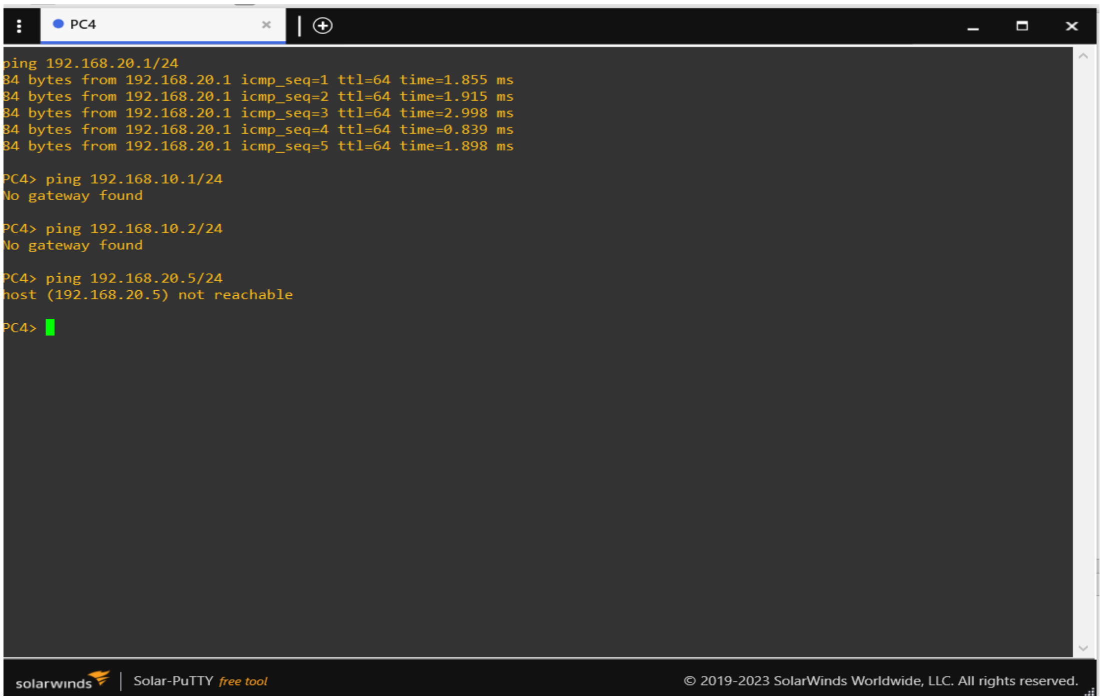
عکس ۱۱ : IP کامپیوتر سوم



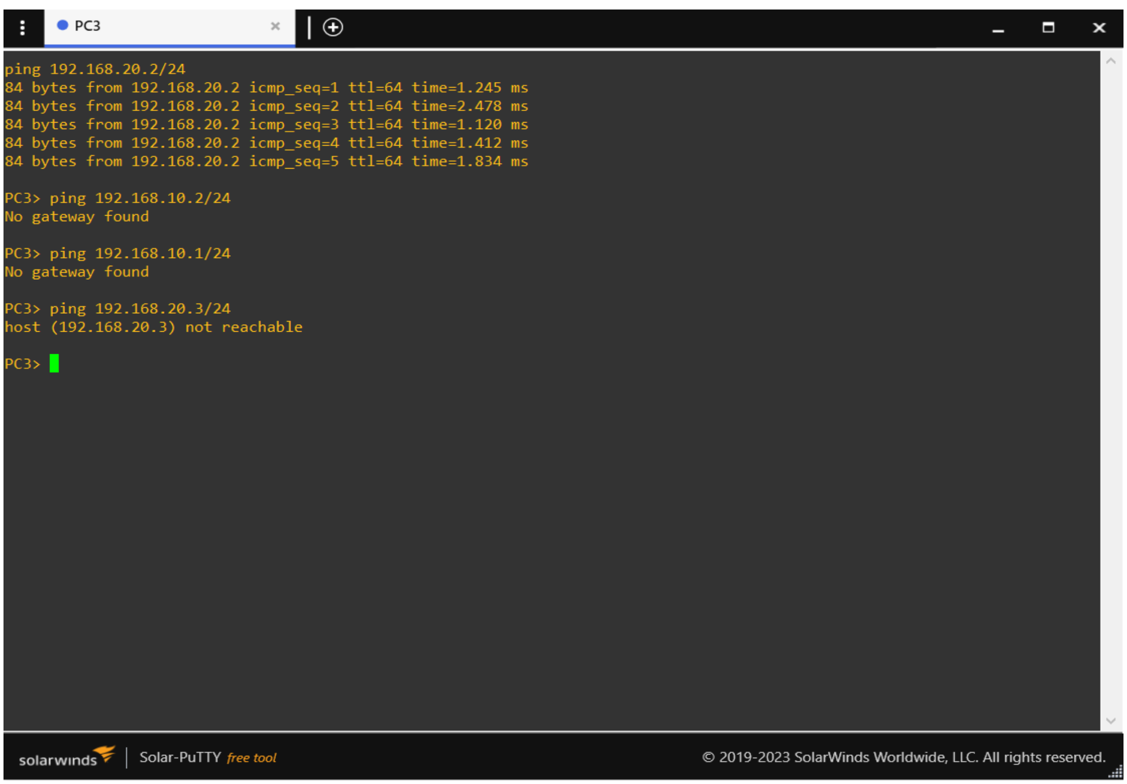
عکس ۱۲ : IP کامپیوتر چهارم

عکس ۱۳ : تست ping از کامپیوتر اول به دومی و سومی و چهارمی و یک IP ست نشده

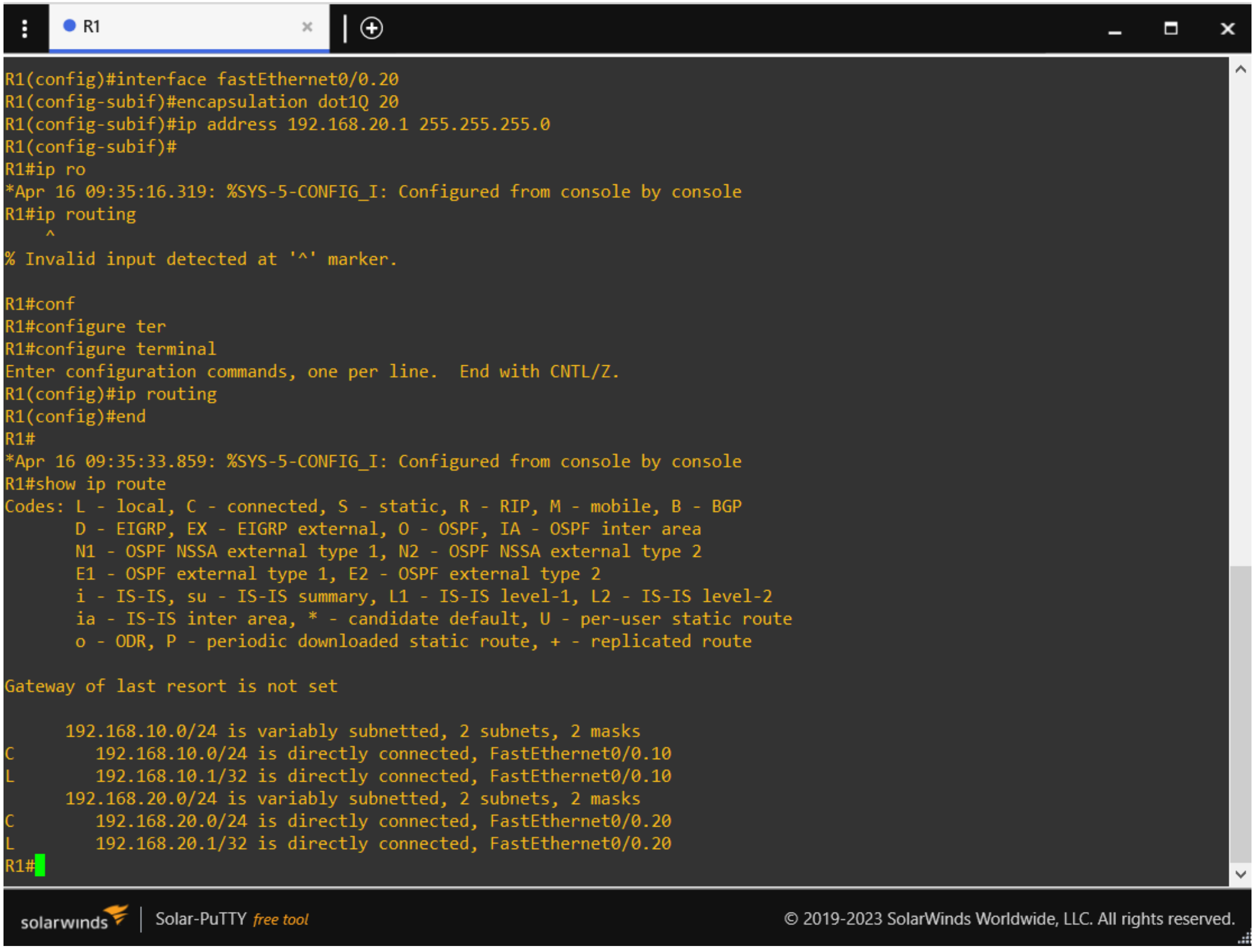
عکس ۱۴ : تست ping از کامپیوتر دومی به اولی و سومی و چهارمی و یک IP ست نشده



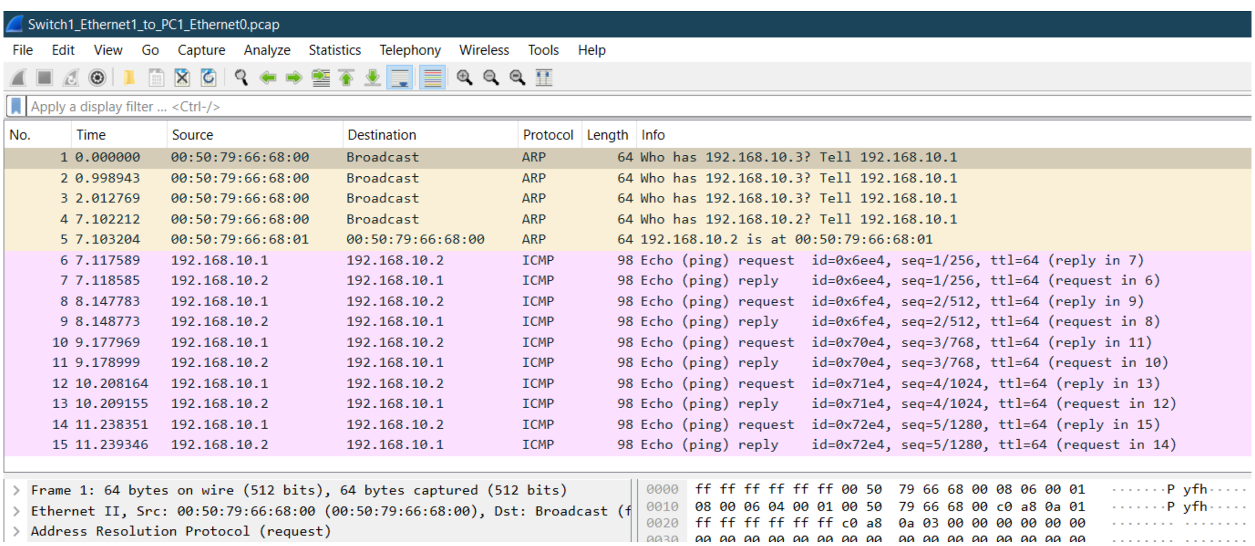
عکس ۱۵ : تست ping از کامپیوتر چهارمی به سومی و اولی و دومی و یک IP ست نشده



عکس ۱۶ : تست ping از کامپیوتر سومی به چهارمی و دومی و اولی و یک IP ست نشده

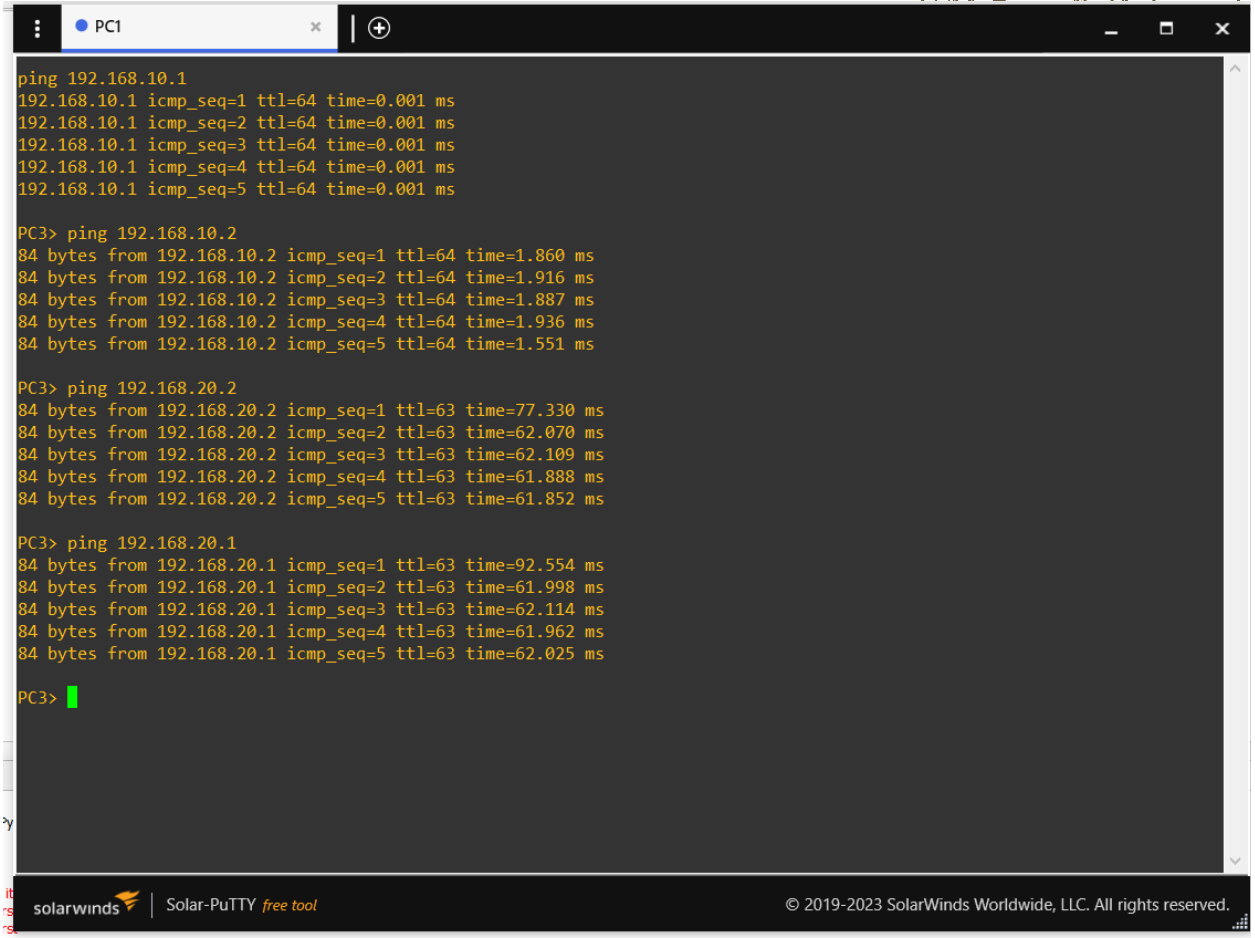


عکس ۱۷ : جدول route روتر



عکس ۱۸ : همان طور که از تصویر پیداست ارتباط را در VLAN 10 با استفاده از wireshark تست نمودیم که پکت های ICMP نشان داده شده حاصل پینگ های گرفته شده است

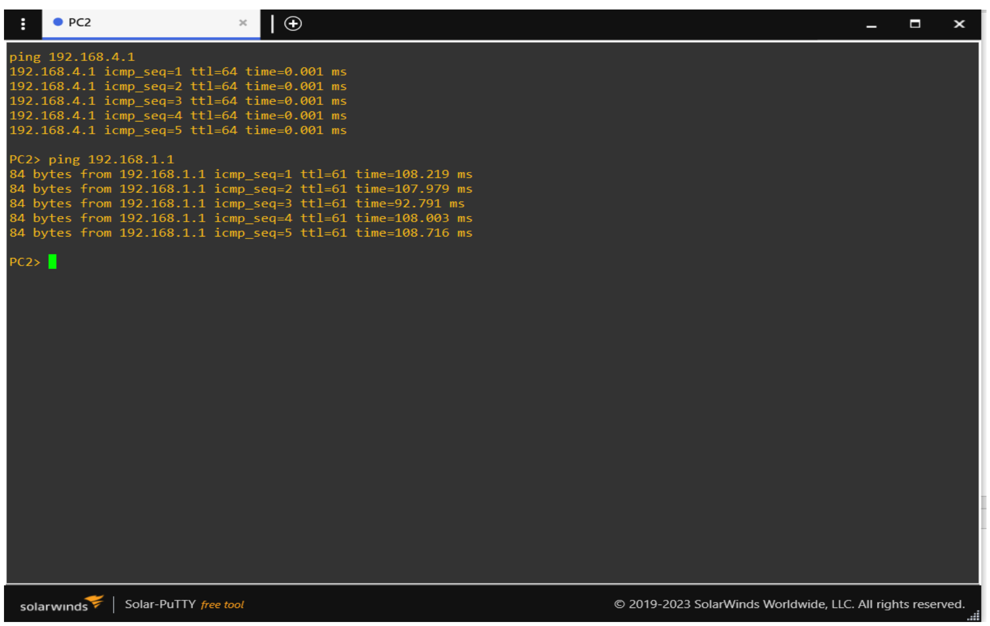
چون ابتدا آی پی های داده شده به کامپیوتر ها gateway نداشت پینگ ها به vlan های دیگر نمیرفت که با دادن gateway در ست شد. نتیجه پینگ های نهایی :



گام سوم

در این بخش ابتدا برای آن که بتوانیم از طریق پورت سریال روتر ها را بهم وصل کنیم به هر کدام یک ماجول PA-4T+ اضافه کردیم. ابتدا در رنج های گفته شده IP دادیم و سپس با استفاده از دستور ip route مسیر ها را برای تک تک روتر ها مشخص کردیم و در نهایت هم ارتباط مابین روتر سمت چپ و پایین را قطع کردیم و دوباره تست گرفتیم. نتایج این بخش:

عکس ۱۹ : ارتباط بین ۲ کامپیوتر



عکس ۲۰ : ارتباط بین ۲ کامپیوتر

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C        10.0.0.0/30 is directly connected, Serial1/0
L        10.0.0.1/32 is directly connected, Serial1/0
S        10.0.0.4/30 [1/0] via 10.0.0.2
C        10.0.0.8/30 is directly connected, Serial1/1
L        10.0.0.9/32 is directly connected, Serial1/1
S        10.0.0.12/30 [1/0] via 10.0.0.2
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, FastEthernet0/0
L        192.168.1.2/32 is directly connected, FastEthernet0/0
S     192.168.4.0/24 [1/0] via 10.0.0.2
R1#
```


عکس ۲۱ : جدول مسیر های روتر اول

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C        10.0.0.0/30 is directly connected, Serial1/0
L        10.0.0.2/32 is directly connected, Serial1/0
C        10.0.0.4/30 is directly connected, Serial1/1
L        10.0.0.5/32 is directly connected, Serial1/1
S        10.0.0.8/30 [1/0] via 10.0.0.6
S        10.0.0.12/30 [1/0] via 10.0.0.6
S     192.168.1.0/24 [1/0] via 10.0.0.1
S     192.168.4.0/24 [1/0] via 10.0.0.6
R2#
```

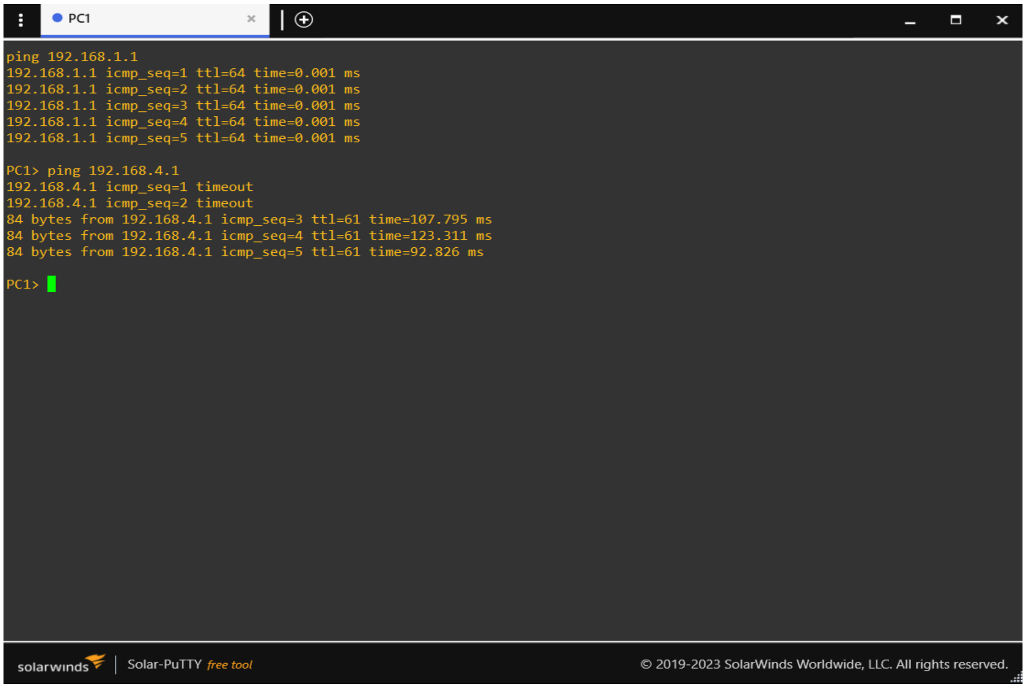

عکس ۲۲ : جدول مسیر های روتر دوم

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
S        10.0.0.0/30 [1/0] via 10.0.0.14
S        10.0.0.4/30 [1/0] via 10.0.0.14
C        10.0.0.8/30 is directly connected, Serial1/0
L        10.0.0.10/32 is directly connected, Serial1/0
C        10.0.0.12/30 is directly connected, Serial1/1
L        10.0.0.13/32 is directly connected, Serial1/1
S     192.168.1.0/24 [1/0] via 10.0.0.14
S     192.168.4.0/24 [1/0] via 10.0.0.14
R3#
```

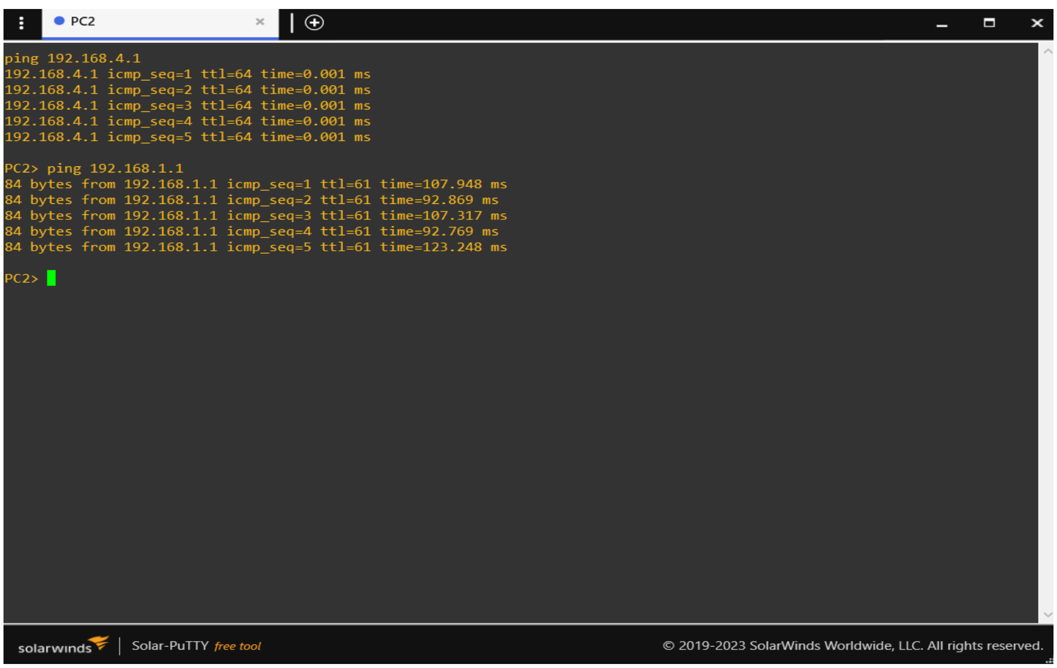

عکس ۲۳ : جدول مسیر های روتر سوم

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
S        10.0.0.0/30 [1/0] via 10.0.0.5
C        10.0.0.4/30 is directly connected, Serial1/0
L        10.0.0.6/32 is directly connected, Serial1/0
S        10.0.0.8/30 [1/0] via 10.0.0.5
C        10.0.0.12/30 is directly connected, Serial1/1
L        10.0.0.14/32 is directly connected, Serial1/1
S     192.168.1.0/24 [1/0] via 10.0.0.5
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, FastEthernet0/0
L        192.168.4.2/32 is directly connected, FastEthernet0/0
R4#
```

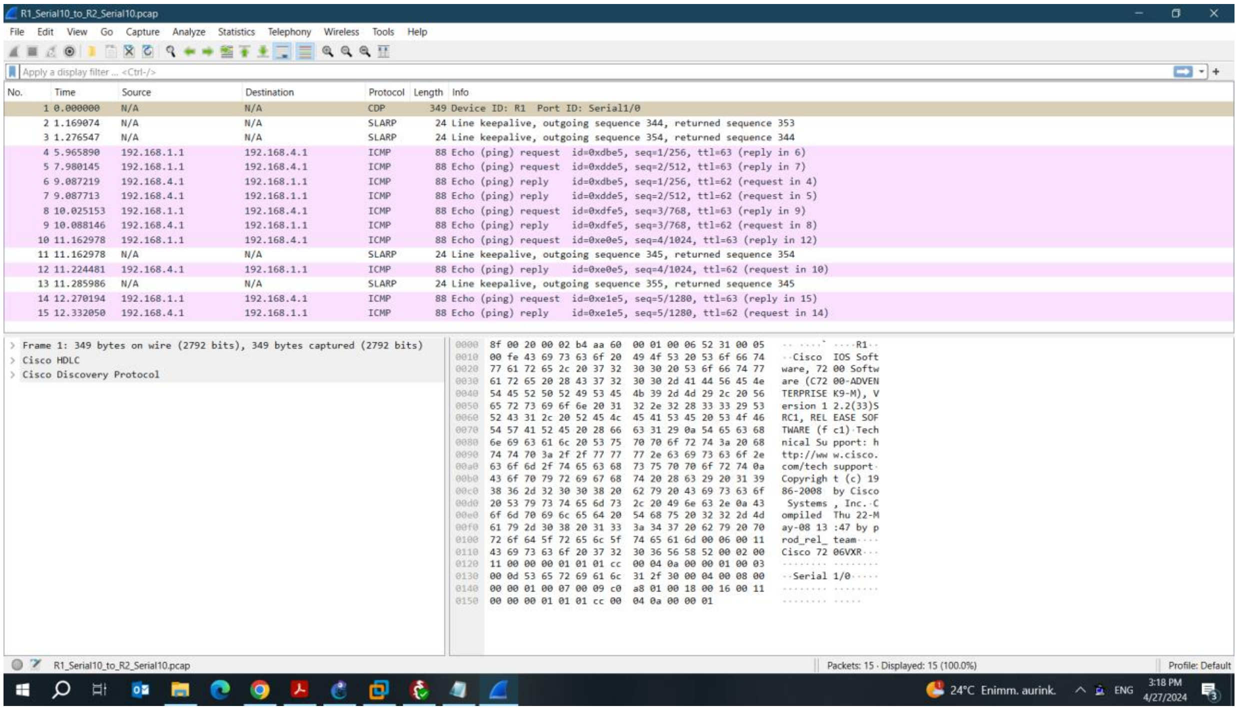

عکس ۲۴ : جدول مسیر های روتر چهارم



عکس ۲۵ : ارتباط بین ۲ کامپیوتر بعد از قطع کردن لینک بین ۲ روتر



عکس ۲۶ : ارتباط بین ۲ کامپیوتر بعد از قطع کردن لینک بین ۲ روتر



عکس ۲۷ : ارتباط بین ۲ کامپیوتر بعد از قطع کردن لینک بین ۲ روتر در وایرشارک

گام چهارم

در این بخش ابتدا برای آن که بتوانیم از طریق پورت سریال روتر ها را بهم وصل کنیم به هر کدام یک ماجول PA-4T+ اضافه کردیم. ابتدا در رنج های گفته شده IP دادیم و سپس با استفاده از دستور router ospf و network مسیر ها را برای تک تک روتر ها مشخص کردیم و در نهایت هم تست گرفتیم. نتایج این بخش:

```
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
O     192.168.3.0/24 [110/129] via 192.168.14.2, 00:04:30, Serial1/1
                     [110/129] via 192.168.12.2, 00:04:30, Serial1/0
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, Serial1/0
L        192.168.12.1/32 is directly connected, Serial1/0
      192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.14.0/24 is directly connected, Serial1/1
L        192.168.14.1/32 is directly connected, Serial1/1
O     192.168.23.0/24 [110/128] via 192.168.12.2, 00:07:03, Serial1/0
O     192.168.34.0/24 [110/128] via 192.168.14.2, 00:07:03, Serial1/1
R3#show ip ospf nei

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.34.1      0   FULL/  -        00:00:36    192.168.14.2    Serial1/1
192.168.23.1      0   FULL/  -        00:00:32    192.168.12.2    Serial1/0
R3#show ip ospf
 Routing Process "ospf 3" with ID 192.168.14.1
 Start time: 01:30:41.436, Time elapsed: 00:17:11.052
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
```

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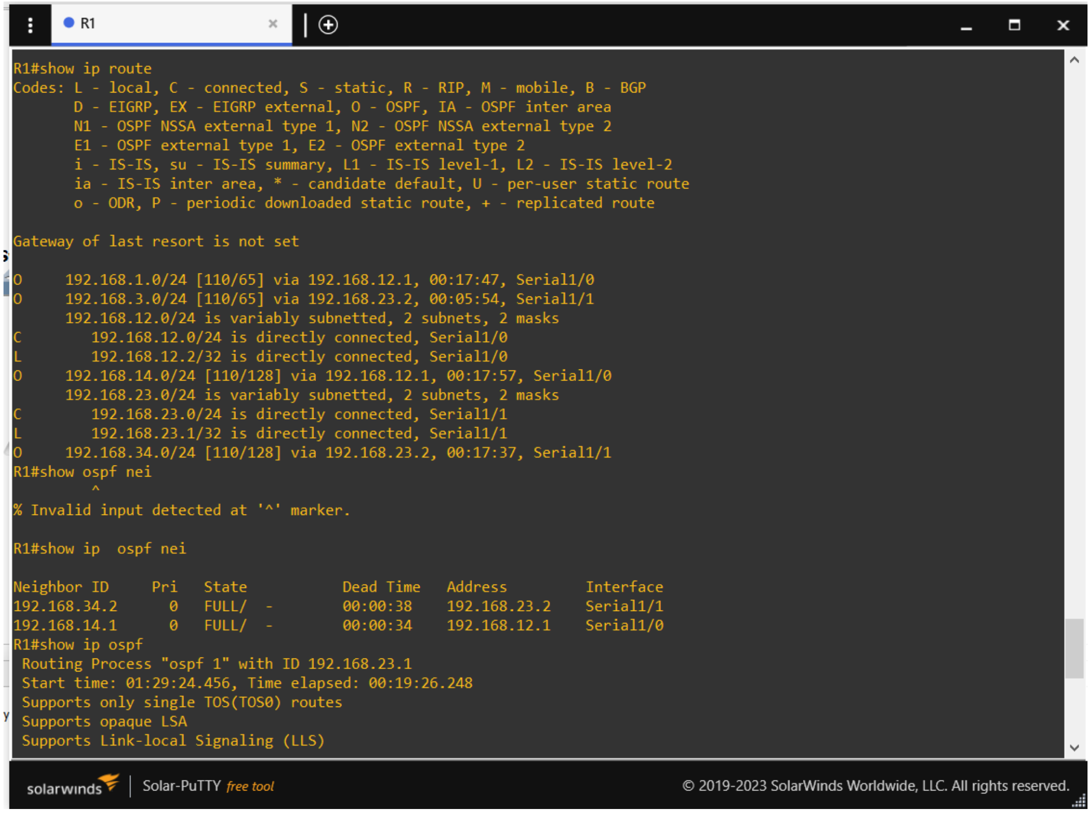
عکس 28 : اطلاعات روتر 3

```
Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPFs 10000 msecs
Maximum wait time between two consecutive SPFs 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
   Area BACKBONE(0)
       Number of interfaces in this area is 3
       Area has no authentication
       SPF algorithm last executed 00:05:05.980 ago
       SPF algorithm executed 7 times
       Area ranges are
       Number of LSA 4. Checksum Sum 0x01C211
       Number of opaque link LSA 0. Checksum Sum 0x000000
       Number of DCbitless LSA 0
       Number of indication LSA 0
       Number of DoNotAge LSA 0
       Flood list length 0
```

عکس 29 : ادامه اطلاعات روتر 3



```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

O     192.168.1.0/24 [110/65] via 192.168.12.1, 00:17:47, Serial1/0
O     192.168.3.0/24 [110/65] via 192.168.23.2, 00:05:54, Serial1/1
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, Serial1/0
L        192.168.12.2/32 is directly connected, Serial1/0
O     192.168.14.0/24 [110/128] via 192.168.12.1, 00:17:57, Serial1/0
      192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.23.0/24 is directly connected, Serial1/1
L        192.168.23.1/32 is directly connected, Serial1/1
O     192.168.34.0/24 [110/128] via 192.168.23.2, 00:17:37, Serial1/1
R1#show ospf nei
         ^
% Invalid input detected at '^' marker.

R1#show ip  ospf nei

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.34.2      0   FULL/  -        00:00:38    192.168.23.2    Serial1/1
192.168.14.1      0   FULL/  -        00:00:34    192.168.12.1    Serial1/0
R1#show ip ospf
 Routing Process "ospf 1" with ID 192.168.23.1
 Start time: 01:29:24.456, Time elapsed: 00:19:26.248
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
```


عکس 30 : اطلاعات روتر 1

```
Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPFs 10000 msecs
Maximum wait time between two consecutive SPFs 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
   Area BACKBONE(0)
       Number of interfaces in this area is 2
       Area has no authentication
       SPF algorithm last executed 00:06:12.348 ago
       SPF algorithm executed 8 times
       Area ranges are
       Number of LSA 4. Checksum Sum 0x01C211
       Number of opaque link LSA 0. Checksum Sum 0x000000
       Number of DCbitless LSA 0
       Number of indication LSA 0
       Number of DoNotAge LSA 0
       Flood list length 0

R1#
```


عکس 31 : ادامه اطلاعات روتر 1

```
R2#
R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

O     192.168.1.0/24 [110/65] via 192.168.14.1, 00:18:28, Serial1/0
O     192.168.3.0/24 [110/65] via 192.168.34.2, 00:06:31, Serial1/1
O     192.168.12.0/24 [110/128] via 192.168.14.1, 00:18:28, Serial1/0
      192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.14.0/24 is directly connected, Serial1/0
L        192.168.14.2/32 is directly connected, Serial1/0
O     192.168.23.0/24 [110/128] via 192.168.34.2, 00:18:13, Serial1/1
      192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.34.0/24 is directly connected, Serial1/1
L        192.168.34.1/32 is directly connected, Serial1/1
R2#show ip ospf nei

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.34.2      0   FULL/  -        00:00:36    192.168.34.2    Serial1/1
192.168.14.1      0   FULL/  -        00:00:33    192.168.14.1    Serial1/0
R2#show ip ospf
 Routing Process "ospf 2" with ID 192.168.34.1
 Start time: 01:30:15.776, Time elapsed: 00:19:11.848
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
 Supports area transit capability
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
```

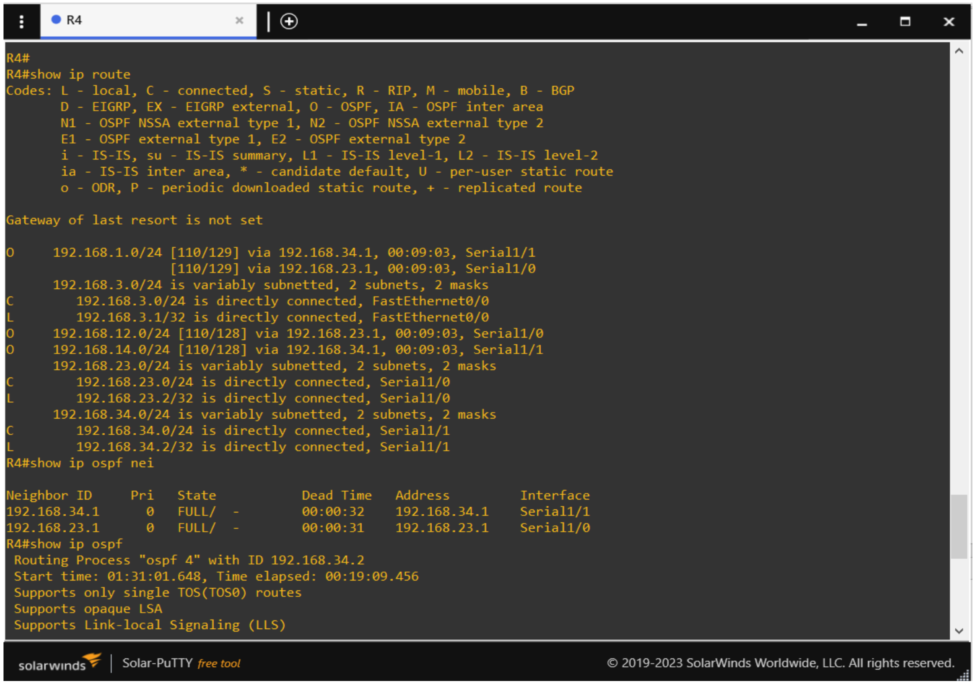
عکس 32 : اطلاعات روتر 2

```
L        192.168.14.2/32 is directly connected, Serial1/0
O     192.168.23.0/24 [110/128] via 192.168.34.2, 00:18:13, Serial1/1
      192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.34.0/24 is directly connected, Serial1/1
L        192.168.34.1/32 is directly connected, Serial1/1
R2#show ip ospf nei

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.34.2      0   FULL/  -        00:00:36    192.168.34.2    Serial1/1
192.168.14.1      0   FULL/  -        00:00:33    192.168.14.1    Serial1/0
R2#show ip ospf
 Routing Process "ospf 2" with ID 192.168.34.1
 Start time: 01:30:15.776, Time elapsed: 00:19:11.848
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
 Supports area transit capability
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Incremental-SPF disabled
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Number of areas transit capable is 0

R2#
```

عکس 33 : ادامه اطلاعات روتر 2



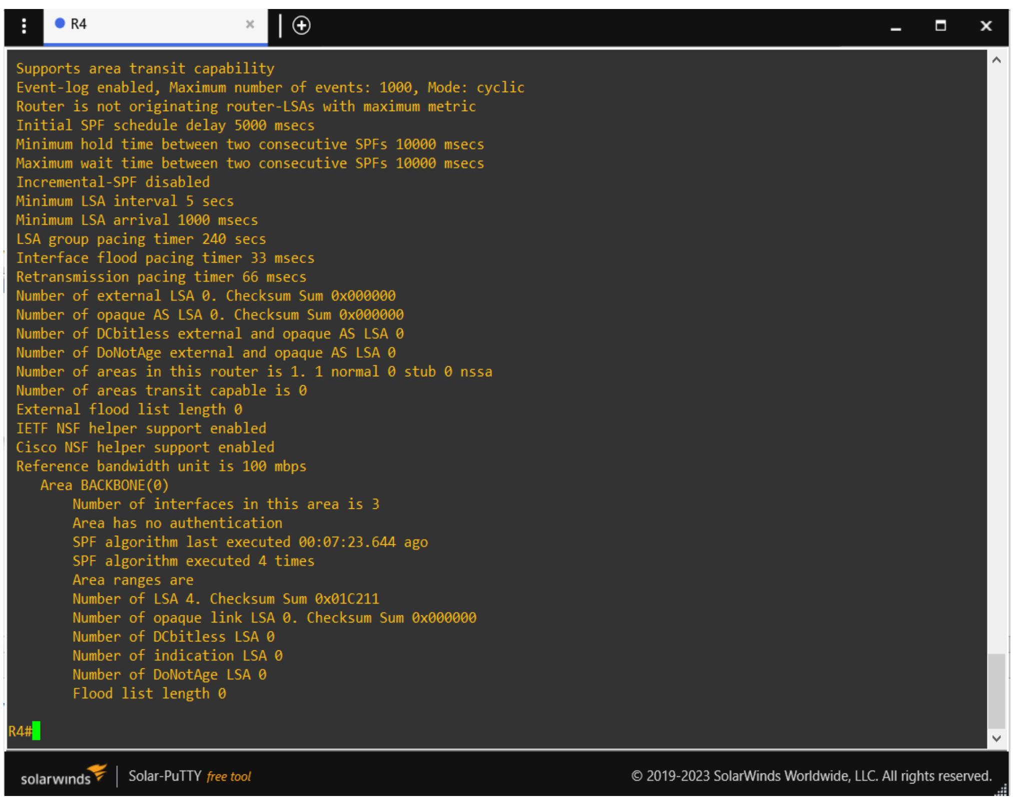

```
R4#
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

O     192.168.1.0/24 [110/129] via 192.168.34.1, 00:09:03, Serial1/1
                     [110/129] via 192.168.23.1, 00:09:03, Serial1/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, FastEthernet0/0
L        192.168.3.1/32 is directly connected, FastEthernet0/0
O     192.168.12.0/24 [110/128] via 192.168.23.1, 00:09:03, Serial1/0
O     192.168.14.0/24 [110/128] via 192.168.34.1, 00:09:03, Serial1/1
      192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.23.0/24 is directly connected, Serial1/0
L        192.168.23.2/32 is directly connected, Serial1/0
      192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.34.0/24 is directly connected, Serial1/1
L        192.168.34.2/32 is directly connected, Serial1/1
R4#show ip ospf nei

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.34.1      0   FULL/  -        00:00:32    192.168.34.1    Serial1/1
192.168.23.1      0   FULL/  -        00:00:31    192.168.23.1    Serial1/0
R4#show ip ospf
 Routing Process "ospf 4" with ID 192.168.34.2
 Start time: 01:31:01.648, Time elapsed: 00:19:09.456
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 Supports Link-local Signaling (LLS)
```

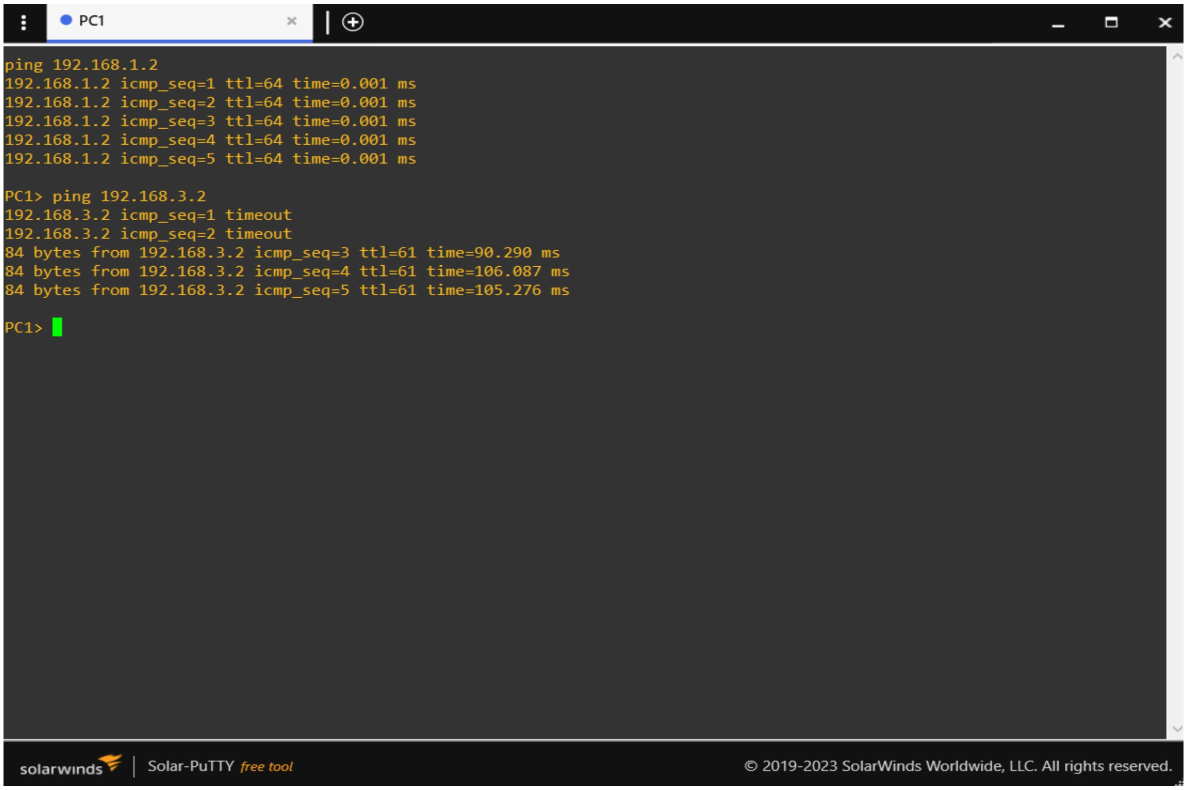
عکس 34 : اطلاعات روتر 4



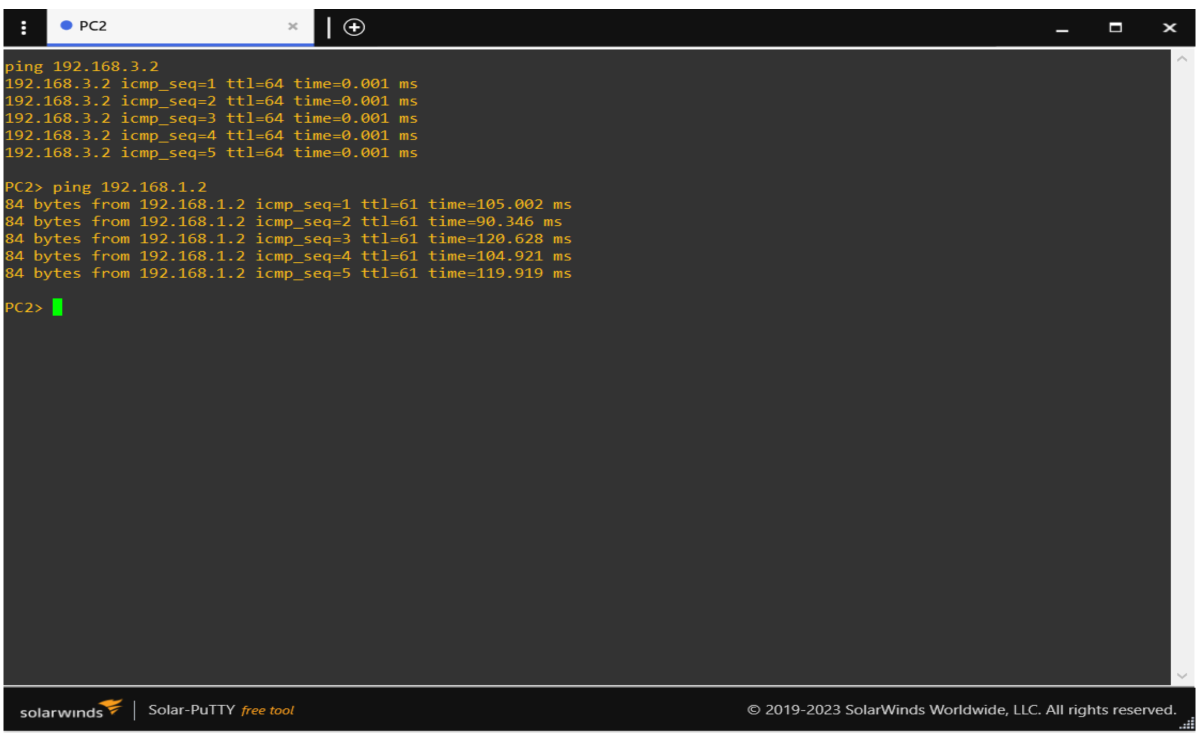

```
 Supports area transit capability
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Incremental-SPF disabled
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Number of areas transit capable is 0
 External flood list length 0
 IETF NSF helper support enabled
 Cisco NSF helper support enabled
 Reference bandwidth unit is 100 mbps
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has no authentication
        SPF algorithm last executed 00:07:23.644 ago
        SPF algorithm executed 4 times
        Area ranges are
        Number of LSA 4. Checksum Sum 0x01C211
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

R4#
```

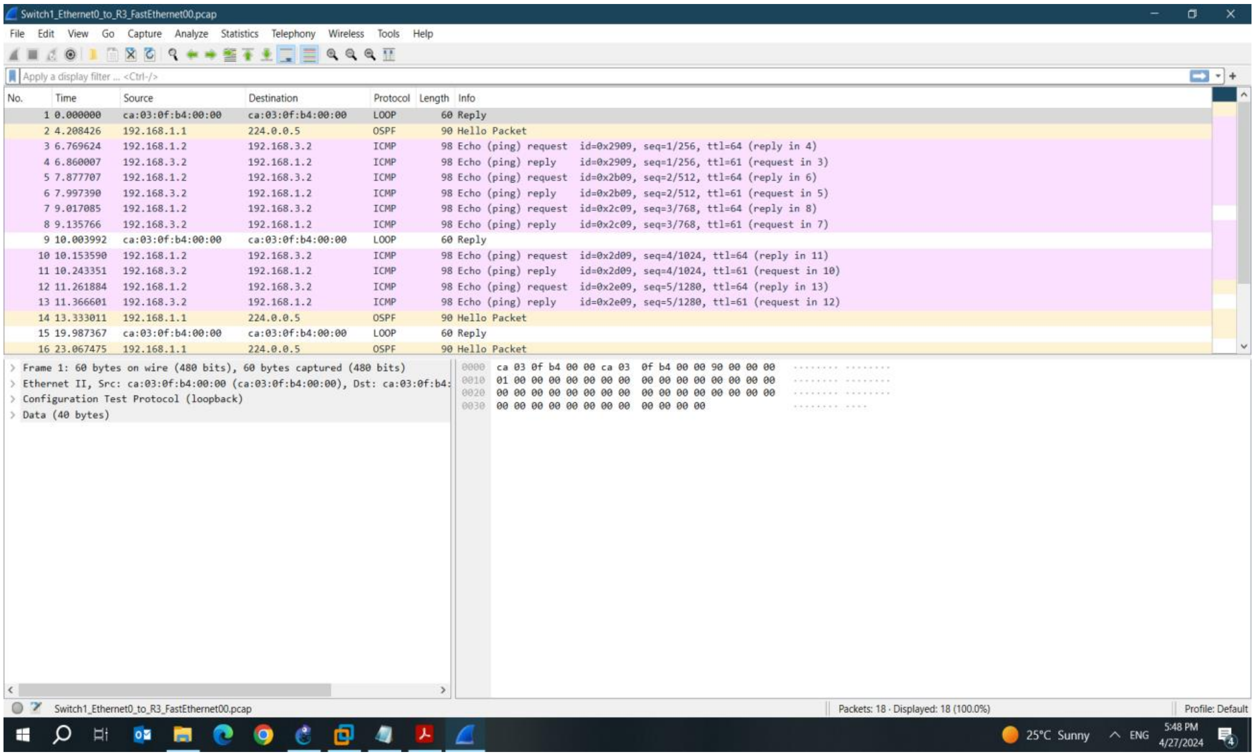
عکس 35 : ادامه اطلاعات روتر 4



عکس 36 : ارتباط بین کامپیوتر اولی به دومی



عکس 37 : ارتباط بین کامپیوتر دومی به اولی



عکس 38 : ارتباط بین دو کامپیوتر در وایرشارک